

Reanalyze Mortiz

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TLDR

The goal is to:

1. Recreate the analysis that Moritz et al. produced for their second experiment, and
2. Perform a different kind of error analysis.

Reading in data from Exp 2 (excluding the line condition):

Repeating Mortiz's analysis

First we read in the relevant data:

```
#point arc
exp2_dotArc <- read.csv(
  here::here("data", "moritz", "exp", "exp2-point-arc.csv"),
  check.names = FALSE
)

#point
exp2_dotX <- read.csv(
  here::here("data", "moritz", "exp", "exp2-point.csv"),
  check.names = FALSE
)

# combine
raw_exp2 <- rbind(exp2_dotArc, exp2_dotX) %>%
  as_tibble(.)

exp2 <- raw_exp2 %>%
  select(assignmentid, mark, flip, variability, error, normalized_error)

exp2 %>% head(5)
```

```
# A tibble: 5 × 6
  assignmentid          mark flip variability    error
normalized_error
```

	<chr>	<chr>	<lgl>	<chr>	<dbl>
<dbl>					
1	3C44YUNSI2PS5YV5YCV0B058A80P...	poin...	FALSE	0,0	0.184
	0.184				
2	3X66WABAJXIOAT0924Z14I54I40G...	poin...	FALSE	0,0.4	0.0925
	0.0925				
3	3LJ7UR74RIDF2H0ZPMHF4767XG6N...	poin...	TRUE	0,0	-0.0672
	0.0672				
4	358UUM7WR03X02AAKRW3G3VRPFL7...	poin...	FALSE	0,0	-0.302
	-0.302				
5	3WQ3B2KGE9GW99EJZET4FEZA47IB...	poin...	TRUE	0,0	0.143
	-0.143				

We also read in the original stimuli, with the line condition filtered out:

```
exp_json_data <- jsonlite::fromJSON(
  here::here("data", "moritz", "stimuli", "exp2-stimuli.json")
  # paste0(getwd(), "/data/moritz/stimuli/exp2-stimuli.json"),
) %>%
  as_tibble(.)

stimuli_df <- exp_json_data %>%
  select(index, seed, noise, flip, type, data) %>%
  mutate(noise = as.character(noise)) %>%
  filter(type != "line")

stimuli_df %>% head(5)
```

```
# A tibble: 5 × 6
  index seed noise      flip type      data
  <int> <int> <chr>    <lgl> <chr>    <list>
1     4     1 c(0, 0)  FALSE point    <df [60 × 2]>
2     5     1 c(0, 0)   TRUE point    <df [60 × 2]>
3     6     1 c(0, 0.4) FALSE point    <df [60 × 2]>
4     7     1 c(0, 0.4)  TRUE point    <df [60 × 2]>
5     8     1 c(0, 0)  FALSE point_arc <df [60 × 2]>
```

Define helper function for plotting¹:

```
plot_results <- function(data, xvar) {
  data %>%
    ggplot(aes(
      y = variability,
      x = {{ xvar }}),
```

¹Borrowed from what the og analysis file had

```

    dist = "norm",
    arg1 = {{ xvar }},
    arg2 = se,
    fill = mark
  )) + #uncertainty
  stat_dist_halfeye(
    show_interval = TRUE,
    position = position_dodge(width = 0.2),
    .width = 0.95,
    size = .5,
    alpha = .8
  ) +
  theme(legend.position = "top") +
  scale_fill_manual(values = c("#76B7B2", "#E15759")) # + # take out
"#59A14F"
}

```

Recreate Moritz et al. plot

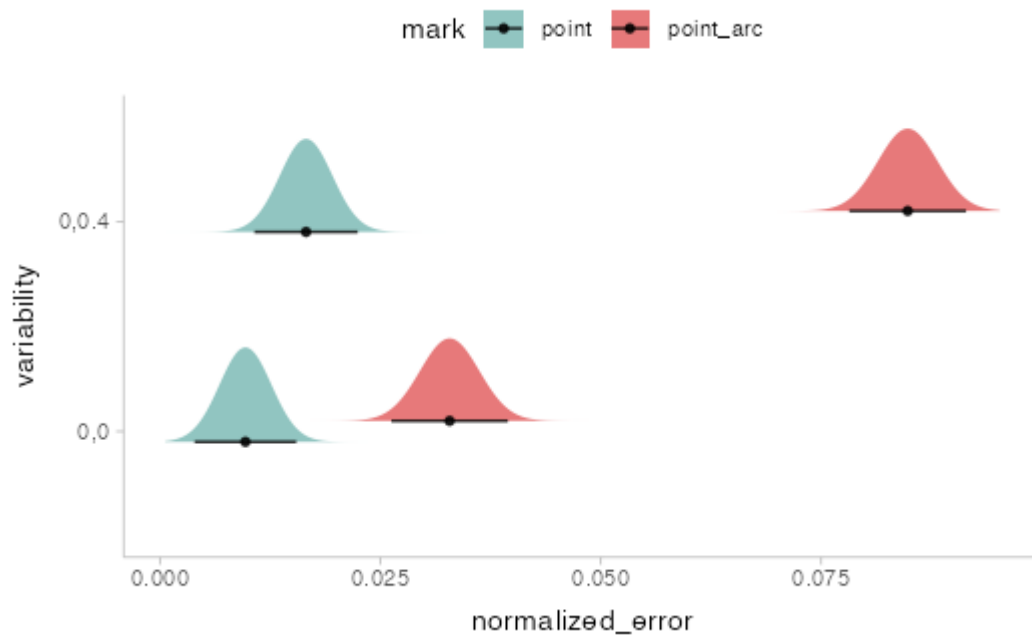
For our own sanity, recreate the plot that Moritz et al. created:

```

p_og <- exp2 %>%
  summarySEwithin(
    data = .,
    measurevar = "normalized_error",
    withinvars = "variability",
    betweenvars = "mark",
    idvar = "assignmentid"
  ) %>%
  plot_results(., normalized_error)

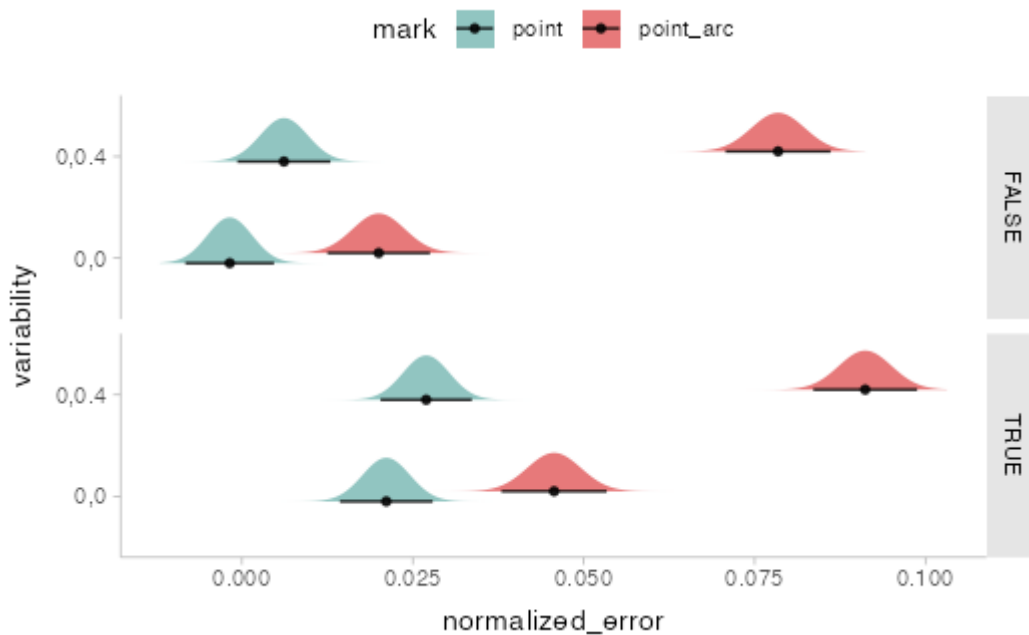
p_og

```



Adding Additional Factors to OG Graph

```
exp2 %>%
  summarySEwithin(
    data = .,
    measurevar = "normalized_error",
    withinvars = c("variability", "flip"), # <- new
    betweenvars = "mark",
    idvar = "assignmentid"
  ) %>%
  plot_results(., normalized_error) +
  facet_grid(flip ~ .)
```



Recalculate error

First, let's see what's the difference between the average data average and the average that Moritz et al. used for their error analysis. Let's first look at how Moritz et al is reporting things in their dataset:

```
raw_exp2 %>%
  select(stimuliindex, seed, mark, flip, variability, average, data_average)
%>%
  group_by(
    stimuliindex,
    seed,
    mark,
    flip,
    variability,
    average,
    data_average
  ) %>%
  summarise(.) %>%
  print(n = 8)
```

```
# A tibble: 96 × 7
# Groups:   stimuliindex, seed, mark, flip, variability, average [96]
  stimuliindex seed mark      flip variability average data_average
      <int> <int> <chr>    <lgl> <chr>          <dbl>      <dbl>
1           4     1 point FALSE  0,0           0.513      0.513
```

2	5	1 point	TRUE	0,0	0.487	0.487
3	6	1 point	FALSE	0,0.4	0.513	0.513
4	7	1 point	TRUE	0,0.4	0.487	0.487
5	8	1 point_arc	FALSE	0,0	0.513	0.495
6	9	1 point_arc	TRUE	0,0	0.487	0.505
7	10	1 point_arc	FALSE	0,0.4	0.513	0.583
8	11	1 point_arc	TRUE	0,0.4	0.487	0.417

i 88 more rows

Observe that for stimuli that share the same seed, the `data_average` is the actual average of the dataset, while the average is the same for stimuli index that share the same seed, flip, and variability².

²for example examine `stimuliindex == 4` and `stimuliindex == 8`

i Verify that data_average is true average

To verify that it is indeed the case the data_average is true average, we can look at the actual stimuli that Moritz et al. provided:

```
ans_df <- stimuli_df %>%
  rename(stimuliindex = index) %>%
  unnest(data) %>%
  group_by(stimuliindex, seed, noise, flip, type) %>%
  rename(mark = type, variability = noise) %>%
  mutate(variability = ifelse(variability == "c(0, 0)", "0,0", "0,0.4"))
%>%
  summarise(mean.y = mean(y))

ans_df
```

```
# A tibble: 96 × 6
# Groups:   stimuliindex, seed, variability, flip [96]
  stimuliindex seed variability flip mark      mean.y
      <int> <int>   <chr>      <lgl> <chr>    <dbl>
1           4     1 0,0        FALSE point    0.513
2           5     1 0,0         TRUE point    0.487
3           6     1 0,0.4      FALSE point    0.513
4           7     1 0,0.4       TRUE point    0.487
5           8     1 0,0        FALSE point_arc 0.495
6           9     1 0,0         TRUE point_arc 0.505
7          10     1 0,0.4      FALSE point_arc 0.583
8          11     1 0,0.4       TRUE point_arc 0.417
9          16     2 0,0        FALSE point    0.429
10         17     2 0,0         TRUE point    0.571
# i 86 more rows
```

Join and check:

```
ans_df %>%
  left_join(
    raw_exp2 %>%
      select(
        stimuliindex,
        seed,
        mark,
        flip,
        variability,
        average,
        data_average
      ) %>%
```

Thus this completes the verification. Indeed data_average corresponds to the actual average.

```
# A tibble: 1 × 2
  seed, data_average` n 7
  <lgl> flip,          <int>
1 TRUE variability, 96
  average,
  data_average
```

This allows us to calculate two kinds of error: (1) reference error, i.e., error that uses the reference average, and (2) actual error, i.e., error that uses the true underlying dataset.

```
exp2 <- raw_exp2 %>%
  select(
    assignmentid,
    mark,
    flip,
    variability,
    reportedaverage,
    average,
    data_average,
    error
  )

exp2 %>% head(5)
```

```
# A tibble: 5 × 8
  assignmentid      mark flip variability reportedaverage average
data_average
  <chr>          <chr> <lgl> <chr>          <dbl>    <dbl>
<dbl>
1 3C44YUNSI2PS5YV5... poin... FALSE 0,0          0.584    0.400
0.439
2 3X66WABAJXIOAT09... poin... FALSE 0,0.4        0.494    0.402
0.483
3 3LJ7UR74RIDF2H0Z... poin... TRUE  0,0          0.438    0.505
0.518
4 358UUM7WR03X02AA... poin... FALSE 0,0          0.0991   0.402
0.424
5 3WQ3B2KGE9GW99EJ... poin... TRUE  0,0          0.709    0.566
0.553
# i 1 more variable: error <dbl>
```

Check to see that the original reported error is obtained via difference between reportedaverage and average:

```
exp2 %>% count(near(reportedaverage - average, error))
```

```
# A tibble: 2 × 2
  `near(reportedaverage - average, error)`    n
  <lgl>                                <int>
1 FALSE                                    2
2 TRUE                                13438
```

Apparently there are two rows where it's not near ...? Let's check:


```
exp2 %>%
  filter(!near(reportedaverage - average, error)) %>%
  mutate(calc_error = reportedaverage - average)
```

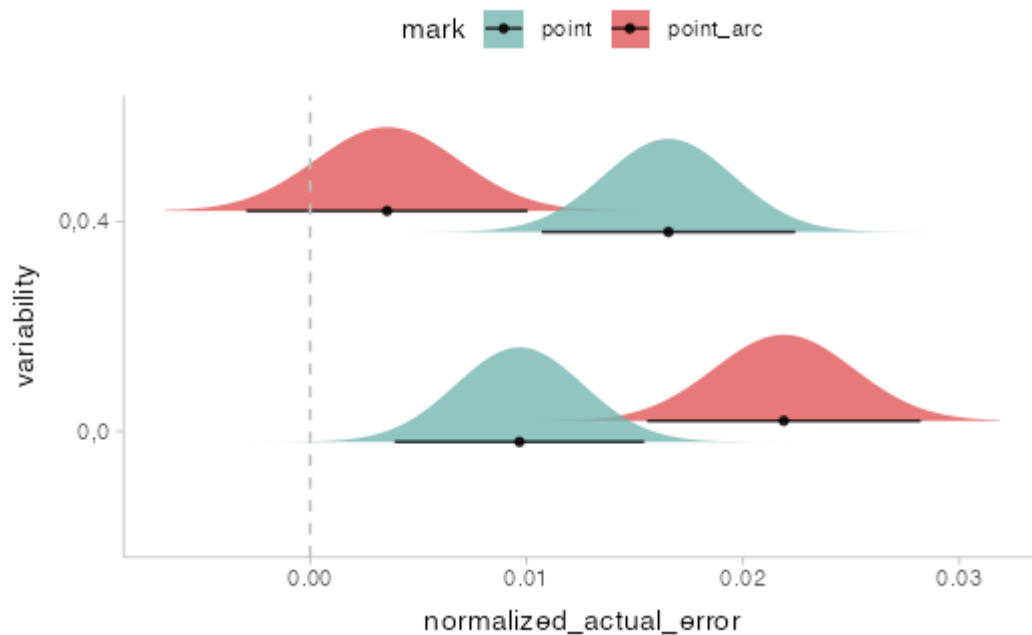
```
# A tibble: 2 × 9
  assignmentid      mark flip variability reportedaverage average
data_average
  <chr>            <chr> <lgl> <chr>                <dbl>    <dbl>
<dbl>
1 34QN5IT0T0RD0MVU... point FALSE 0,0                0.257    0.257
0.257
2 3GD6L00D3TXSS0IG... point TRUE  0,0.4            0.566    0.566
0.566
# i 2 more variables: error <dbl>, calc_error <dbl>
```

This appears to just be due to rounding error. So this confirms that indeed $\text{error} = \text{reportedaverage} - \text{average}$.

Reanalyze with different error definition

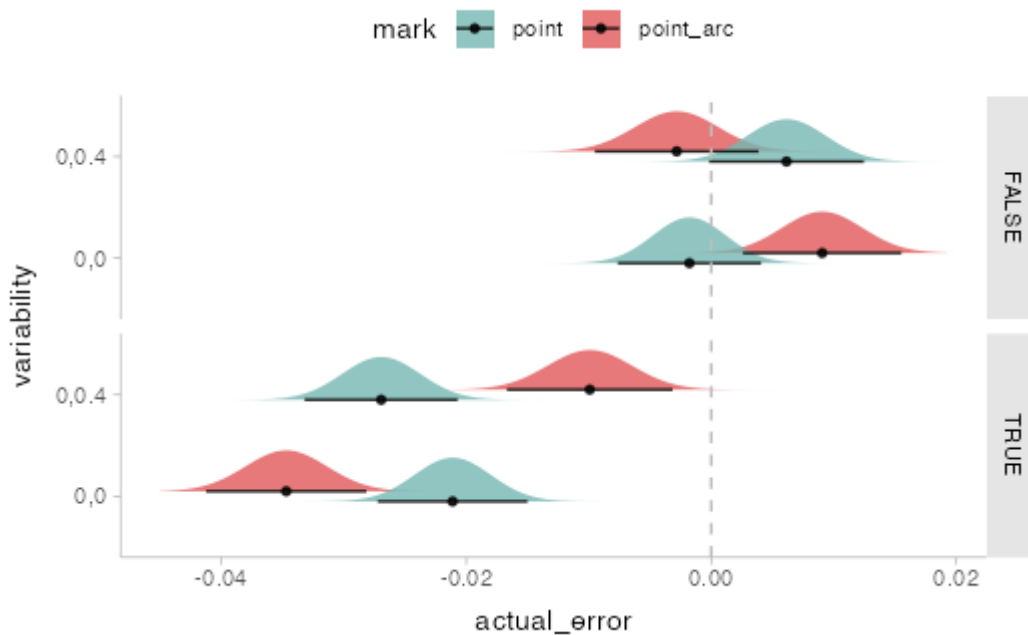
Using the exact same calculations, we obtain the following:

```
exp2 %>%
  rename(ref_error = error) %>%
  mutate(
    actual_error = reportedaverage - data_average,
    normalized_actual_error = ifelse(flip == TRUE, -actual_error,
actual_error)
  ) %>%
  summarySEwithin(
    data = .,
    measurevar = "normalized_actual_error",
    withinvars = "variability",
    betweenvars = "mark",
    idvar = "assignmentid"
  ) %>%
  plot_results(., normalized_actual_error) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray")
```



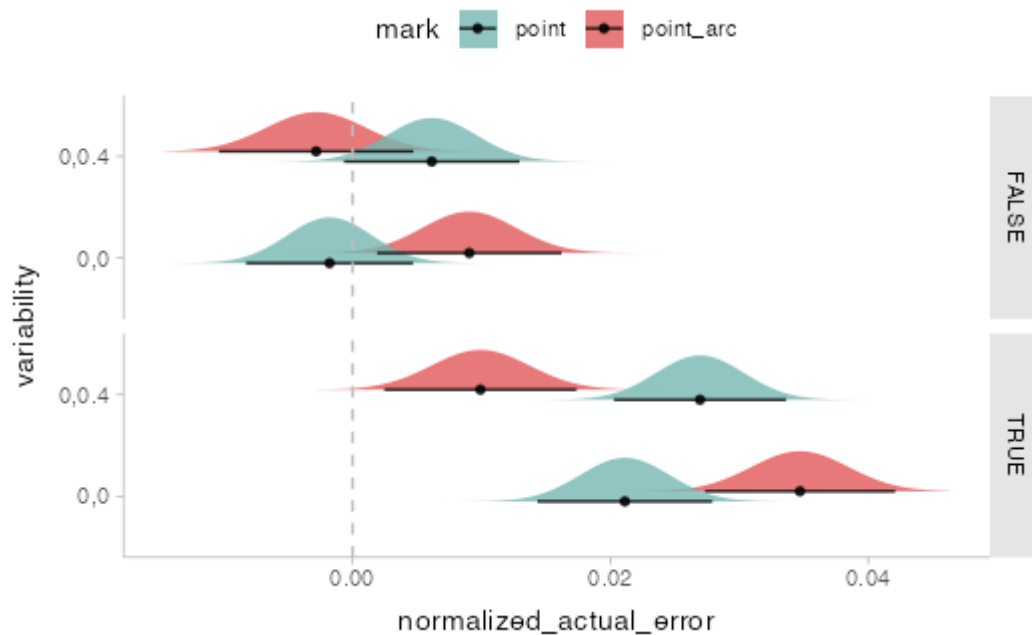
If we don't use the "normalization", and facet by the factor of flip, then we get this:

```
exp2 %>%
  rename(ref_error = error) %>%
  mutate(
    actual_error = reportedaverage - data_average,
    normalized_actual_error = ifelse(flip == TRUE, -actual_error,
actual_error)
  ) %>%
  summarySEwithin(
    data = .,
    measurevar = "actual_error",
    withinvars = c("variability", "flip"),
    betweenvars = "mark",
    idvar = "assignmentid"
  ) %>%
  plot_results(., actual_error) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray") +
  facet_grid(flip ~ .)
```



We can also make the direction look nicer and easier to read:

```
exp2 %>%
  rename(ref_error = error) %>%
  mutate(
    actual_error = reportedaverage - data_average,
    normalized_actual_error = ifelse(flip == TRUE, -actual_error,
actual_error)
  ) %>%
  summarySEwithin(
    data = .,
    measurevar = "normalized_actual_error",
    withinvars = c("variability", "flip"),
    betweenvars = "mark",
    idvar = "assignmentid"
  ) %>%
  plot_results(., normalized_actual_error) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray") +
  facet_grid(flip ~ .)
```



It seems to me that there are several things that are unexplained in the original analysis:

- When the original plot is flipped (when `flip == TRUE`), there seems to be a bigger bias compared to the case when it's not flipped.
- When calculating error using the true mean, instead of the mean for the plot with the same seed, we don't see the same trend that more variability implies bigger error.

What is the relationship between actual mean and reference mean, vs the plot type?

As the title states, for the plots that share the same reference line, does being pointArc imply more variability?

```
s_df <- raw_exp2 %>%
  select(
    stimuliindex,
    seed,
    mark,
    flip,
    variability,
    average,
    data_average
  ) %>%
  group_by(
    stimuliindex,
    seed,
```

```

    mark,
    flip,
    variability,
    average,
    data_average
  ) %>%
  summarise(.)

s_df %>% head(5)

```

```

# A tibble: 5 × 7
# Groups:   stimuliindex, seed, mark, flip, variability, average [5]
  stimuliindex seed mark      flip variability average data_average
      <int> <int> <chr>    <lgl> <chr>          <dbl>      <dbl>
1           4     1 point FALSE  0,0           0.513      0.513
2           5     1 point TRUE   0,0           0.487      0.487
3           6     1 point FALSE  0,0.4       0.513      0.513
4           7     1 point TRUE   0,0.4       0.487      0.487
5           8     1 point_arc FALSE  0,0           0.513      0.495

```

Now let's plot the density and mean:

```

means_df <- s_df %>%
  group_by(flip, variability, mark) %>%
  summarize(mean_val = mean(data_average), .groups = "drop")

s_df %>%
  ggplot(aes(x = data_average, color = mark)) +
  geom_density() +
  geom_vline(
    data = means_df,
    aes(xintercept = mean_val, color = mark),
    linetype = "dashed"
  ) +
  facet_wrap(flip + variability ~ .)

s_df %>%
  ggplot(aes(x = data_average, color = mark)) +
  geom_density() +
  stat_pointinterval(
    aes(y = -0.05),
    point_interval = mean_qi,
    position = position_dodge(width = 0.4)
  ) +
  facet_wrap(flip + variability ~ .)

```

